

ABDUL RAFFY

B.S. Artificial Intelligence | AI/ML & Research

Email: abdulraffy0428@gmail.com
Phone: +92 3457252529
Country: Pakistan
GitHub: github.com/Raffy0-1
Portfolio: abdul-raffy-portfolio.vercel.app

Academic Profile

B.S. Artificial Intelligence student with practical experience in machine learning, Natural Language Processing (NLP), LLM pipelines, and interdisciplinary computational research. Progressed from top-ranked AI Research Intern to AI Researcher at AIC, contributing to interdisciplinary scientific and cognitive modeling projects. Currently exploring physics-informed machine learning frameworks for complex scientific systems. Seeking Master's-level study in Artificial Intelligence at a leading Chinese university to expand technical foundations and research potential under academic supervision.

Research Experience

AIC (AI Cadmey) *Remote / Hybrid*
AI Researcher 2024 – Present

- Appointed as AI Researcher following top-ranked internship performance; participate in interdisciplinary research applying machine learning methodologies across scientific, temporal, and cognitive intelligence domains.
- Collaborate in a team environment on foundational computational research projects, including:
  - **Alpha-Centauri:** Trajectory modeling and planetary habitability parameter analysis.
  - **Basal Ganglia:** Neuromorphic computational modeling exploring cognitive mechanisms and decision pathways.
  - **HOS, Raymax, & Harmonic Origin:** Signal modeling, mathematical frameworks, and data synthesis.
- Execute computational experimentation, analyze dataset outputs, and assist in evaluating model behavioral consistency.

AIC (AI Cadmey) *Remote / Hybrid*
AI Research Intern 2024 (4 Months)

- Participated in a structured 4-month AI research training program focusing on core machine learning concepts, Python experimentation, and scientific literature analysis.
- Led the assigned intern cohort and secured the **#1 Rank** in final evaluation benchmarks, leading to appointment as an AI Researcher.

Ongoing Research Project

Physics-Informed AI for Early-Universe Dynamics *Independent Research Project*
Undergraduate Researcher 2026 – Present

- Investigating physics-informed machine learning (PINN) frameworks for modeling cosmological evolution using simulated numerical data and mathematical physics constraints.
- Incorporating differential equation constraints into neural network loss functions to reduce black-box predictions and promote physical consistency.

- Evaluating model validation metrics focusing on prediction error, constraint satisfaction, domain shift stability, and interpretability.

Education

The Islamia University of Bahawalpur *Bahawalpur, Pakistan*
Bachelor of Science in Artificial Intelligence (BS AI) 2023 – 2027 (Expected)

- **Relevant Coursework:** Machine Learning, Deep Learning, Natural Language Processing, Data Structures & Algorithms, Linear Algebra, Probability & Statistics, Database Management Systems, Object-Oriented Programming (Python/C++).
- Focused academic concentration on artificial intelligence architectures, predictive modeling, and computational research methods.

Allama Iqbal Higher Secondary School *Shujabad, Pakistan*
Higher Secondary School Certificate (HSSC / F.Sc. Pre-Engineering) 2021 – 2023

- Academic focus on Mathematics, Physics, and Chemistry.

Allama Iqbal Higher Secondary School *Shujabad, Pakistan*
Secondary School Certificate (SSC / Matriculation Science) 2019 – 2021

Engineering Experience

DevelopersHub Corporation *Islamabad, Pakistan (Remote)*
AI/ML Engineer Intern 2026

- Implemented practical machine learning pipelines across two engineering development phases.
- **Phase 1 (Predictive Engineering):** Built financial and real estate predictive models using regression, decision trees, and ensemble algorithms for Stock Price Prediction and House Price Forecasting.
- **Phase 2 (Advanced NLP & RAG):** Developed a Context-Aware Retrieval-Augmented Generation (RAG) Chatbot utilizing LangChain, vector embeddings, and LLM orchestration. Constructed an end-to-end Customer Churn Prediction ML Pipeline with data preprocessing and evaluation metrics.

Legal Firm (UK-Based) *Bahawalpur, Pakistan*
Junior Conveyancing Documentation Assistant 2023 (2 Months)

- Assisted with structured legal documentation and record organization, developing rigorous attention to detail and data management skills.

Selected AI & Computational Projects

Quantifying Nostalgia: Large-Scale NLP Sentiment Study *GitHub: Raffy0-1/Quantifying-Nostalgia*
Independent AI Project 2024 – 2025

- Executed a large-scale Natural Language Processing study on a dataset of approximately **930,000 movie reviews** to quantify nostalgic discourse patterns.
- Implemented text normalization, tokenization, stop-word removal, and TF-IDF feature extraction pipelines.
- Benchmarked classification performance across Logistic Regression, Linear SVM, Random Forest, and Naive Bayes classifiers to isolate temporal sentiment shifts.

Stellar Pathways: Journey to Alpha Centauri *GitHub: Raffy0-1/Stellar-Pathways*
AIC Interdisciplinary Project 2024 – 2025

- Modeled propulsion parameters and interplanetary trajectories toward the Alpha Centauri system (Proxima Centauri b).
- Conducted computational analysis on energy efficiency and travel duration constraints for long-duration space exploration scenarios.

Context-Aware RAG System & Vector Retrieval *DevelopersHub Project*
AI Engineering Project 2026

- Constructed an interactive document Q&A pipeline leveraging document chunking strategies, vector database indexing, and semantic similarity search via LangChain.
- Enforced contextual retrieval bounds prior to response generation to mitigate output hallucination.

Supervised ML Classification Implementations *Independent ML Projects*
Machine Learning Projects 2023 – 2024

- **Review Manipulation Detection:** Developed supervised ML models to identify deceptive patterns in online customer reviews.
- **SONAR Target Classification:** Trained binary classification models (Logistic Regression, KNN) on SONAR frequency data to differentiate seabed rocks from metallic mines.

Technical Skills

- **Programming Languages:** Python, C++, SQL, JavaScript, HTML5/CSS3.
- **Machine Learning & AI:** Supervised Learning, Regression, Classification, Random Forests, SVM, Feature Engineering, Model Evaluation Metrics, Physics-Informed ML Concepts.
- **NLP & Deep Learning:** Natural Language Processing, TF-IDF, NLTK, Retrieval-Augmented Generation (RAG), LangChain, LLM API Integration.
- **Data Science & Visualization:** Pandas, NumPy, Scikit-Learn, Matplotlib, Seaborn, Statistical Analysis, Data Preprocessing.
- **Developer Tools & Environment:** Git, GitHub, VS Code, Jupyter Notebooks, Linux Environment.

Research Interests

- **Artificial Intelligence & Machine Learning:** Supervised & Unsupervised Learning, Predictive Modeling.
- **Scientific Machine Learning:** Physics-Informed Neural Networks (PINNs), AI for Scientific Simulation.
- **Natural Language Processing & LLMs:** Retrieval-Augmented Generation, Text Analytics, Semantic Representation.
- **Intelligent Systems:** Interdisciplinary AI Applications and Computational Intelligence.

Honors & Achievements

- **Rank #1 in AIC Research Internship (2024):** Achieved top position out of the intern cohort based on research performance, problem-solving evaluation, and technical execution.
- **Advancement to AI Researcher (2024):** Appointed to AI Researcher role at AIC following internship completion.
- **Sustained Research Contribution:** Selected to collaborate on 5+ interdisciplinary computational projects at AIC.

Languages

- **English:** Full Professional Proficiency (Academic & Technical Communication).
- **Urdu:** Native Speaker.
- **Chinese:** Elementary Proficiency (HSK 1 Level).

References

Academic and professional references available upon request.